Walks, Paths and Connectivity from the Adjacency Matrix

A study note on what $(A^k)_{ij}$ counts and how A shows whether a graph is connected

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1 Introduction

1.1 The questions and where they come from

This note starts from a lecture whiteboard, photographed twice from slightly different angles [1, 2]. The board shows a graph with N vertices that is given through its adjacency matrix A, and it asks two questions about A [1]. The first question is what the entry (i, j) of a power of A represents. The second question is how to know from A whether the graph is connected. The board then defines a connected graph as one in which, for any two vertices, there exists a path between them [1]. It does not say what a path is. The assignment text that came with the photos adds a third question: what is the definition of a path between two vertices n_i and n_j? The assignment text also states which meaning of the word path the course uses; Convention 2.5 records it. We call the three questions Q1, Q2 and Q3:

Q1 What does the entry $(A^k)_{ij}$ represent?

Q2 How can one determine from A whether the graph is connected?

Q3 What is the definition of a path between n_i and n_j?

The board also holds a small drawing of three vertices and a 3×3 matrix. We use that matrix as the main example, and Section 5.3 reports where the drawing and the matrix differ.

1.2 The story used throughout

In plain words. Think of a group of towns. Some pairs of towns have a road between them. Every road can be driven in both directions. A trip follows roads, one after another. You may drive past the same town again, or the same road again. Two questions come up. How many different trips of a given size lead from one town to another? And can you get from every town to every other town?

Table 1 fixes the story words once. The plain words paragraphs in this note use only the story words in the right column. The technical terms in the left column are defined where they first appear.

1.3 What the note shows

The short answers are as follows. The sections named give the definitions and proofs.

- **Q1.** For $k \geq 1$, the entry $(A^k)_{ij}$ is the number of *walks* of length k from n_i to n_j (Theorem 5.1). It counts walks, not paths; Section 5.4 shows by example why the difference matters.

Table 1: The story dictionary. One story word for each technical term.

Technical term	Story word
vertex	town
edge	road (it can be driven both ways)
adjacent vertices	neighbouring towns (a road joins them)
degree of a vertex	number of roads at a town
adjacency matrix	the road table
walk	trip
length of a walk	number of drives (a road counts every time it is driven)
closed walk	round trip
trail	trip that never drives a road twice
path (course meaning)	tidy route (never visits a town twice)
cycle	loop trip (a round trip that uses no road twice)
distance	fewest drives between two towns
connected graph	every town can be reached from every other town
connectivity component	island (a group of towns that reach one another)

- **Q2.** The graph on $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and this holds if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 6.10).
- **Q3.** Under the course convention, a path between n_i and n_j is a sequence of pairwise distinct vertices that starts at n_i, ends at n_j, and has adjacent vertices in each consecutive pair (Definition 3.35).

The course textbook defines the graph, the adjacency matrix, walks, paths, cycles, distance and connectivity [3]. As far as a text search of the book for the adjacency matrix shows, it mentions that matrix on the table of contents and on printed pages 7, 11, 12, 34 and 61, and the pages we viewed contain neither a theorem that counts walks with matrix powers nor a matrix test for connectivity [3]. We did not read the whole book, so we state this only as the result of that search. Both results are therefore proved here in full. Lecture notes state them as well [4, 5, 6], and we name the source beside each statement.

1.4 How the note is organized

Section 2 fixes notation and the conventions we adopt where the board, the assignment text and the textbook leave a choice. Section 3 gives the definitions in the textbook's order, each followed by a remark, an example and a non-example. Section 4 answers Q3 and proves the lemmas about walks and paths that the other answers need. Section 5 answers Q1. Section 6 answers Q2 and works two examples. Section 7 states limits, and Section 8 gives the direct answers. Appendix A lists the sources, the pages viewed and how to rerun every computation.

Statements are marked by their form, not by labels. A definition carries the textbook's number where it has one. A theorem, lemma or proposition that has a proof printed here is our own proof; a citation in its first sentence says which source also states the result. A reading of the photos or of the assignment text is a numbered convention in Section 2.

2 Notation and conventions

2.1 Notation

Let $G = (V, E)$ be a graph, with vertex set V and edge set E (the exact definition is Definition 3.1). We use the following letters throughout.

- $V = \{n_1, \ldots, n_N\}$ is the set of vertices and $N = |V|$ is their number. The board names the vertices $n_1, \ldots, n_N$ and writes N [1]. The textbook writes $v_1, \ldots, v_n$ [3, p. 11]; we do not use that form again.
- e_{ij} is the edge between n_i and n_j. In a simple graph e_{ij} and e_{ji} are the same edge.
- $A = (A_{ij})$ is the adjacency matrix, an $N \times N$ matrix of zeros and ones (Definition 3.9). Its power A^k is the product of k copies of A, with $A^1 = A$, and $A^0 = I$ is the identity matrix, which has 1 on the diagonal and 0 elsewhere.
- k is a nonnegative integer used for the length of a walk and for the exponent of A.
- $d(n_i)$ is the degree of n_i (Definition 3.13).
- u, v, w are arbitrary vertices.
- For $N \times N$ matrices X and Y the entry $(XY)_{ij}$ equals $\sum_{r=1}^{N} X_{ir} Y_{rj}$.

2.2 Conventions

The photos, the assignment text and the textbook leave several points open. Each choice we make is listed here and cited by number afterwards.

Convention 2.1 (kind of graph)**.** All graphs are finite, simple and undirected. Simple means no loops and no repeated edges. The assignment text states this, and the symmetric matrix with zero diagonal on the board is consistent with it [1]. Directed graphs and graphs with repeated edges are outside the scope of this note; the adjacency matrix of a directed graph is not symmetric in general, so the statements below that use symmetry do not carry over unchanged.

Convention 2.2 (the mark above A)**.** The board writes a small mark above A in Q1. In the two photos it looks like the letter n or k set between quotation-like marks, so the board alone does not settle the reading [1, 2]. The assignment text asks about A^k, so we read the mark as an exponent k. A prime or quotation marks would ask about another matrix or only name A.

Convention 2.3 (the board's letters)**.** The board writes (G, V) for the graph and "$e_{ij} \in V$" in the rule that defines A_{ij} [1, 2]. The textbook writes $G = (V, E)$ and the same rule with $e_{ij} \in E$ [3, p. 9]. We read the board's V in that rule as the edge set E. This is a difference of notation, and we write E.

Convention 2.4 (matrix against drawing)**.** The board's drawing and the board's matrix disagree, edge by edge (Section 5.3). The problem statement on the board says the graph is represented through its adjacency matrix [1], so the matrix is the example graph. We read the arrowhead on one line of the drawing as a stray mark, because the matrix is symmetric and the graph is undirected.

Convention 2.5 (the word "path")**.** The board does not define path [1]. The assignment text says that in the course a path is a walk with no repeated vertex. The textbook has the same object under the name *simple path* (Definition 1.2.13), and it uses the word *path* for a walk with no repeated edge (Definition 1.2.12) [3, p. 22]. Because the two meanings of "path" differ, we write *trail* for the textbook's Definition 1.2.12 and say "the textbook's path" when we quote it. Plain "path" always means no repeated vertex. Table 2 compares the names.

Convention 2.6 (walks of length zero)**.** A single vertex is a walk of length 0 from that vertex to itself. The textbook does not say whether length 0 is allowed in its walk definition on the pages we viewed (printed pp. 21 to 23) [3]. Grinberg allows it [7, p. 36, Definition 2.9.1(a)]. We adopt it as a convention. It makes $A^0 = I$ count walks of length 0, and it makes the walk-to-path lemma (Lemma 4.2) true for $u = v$.

Convention 2.7 (page numbers)**.** Textbook page numbers are the printed numbers; the page of the PDF file is one higher. The textbook is in Armenian; we give its terms in English translation, and every definition is cited by the book's number.

This note makes no claim about what the lecturer intended beyond what the photos and the assignment text show.

3 Background: graphs, matrices, walks and paths

Let $G = (V, E)$ be a graph, $u, v \in V$ and $e \in E$. The definitions below follow the textbook's order [3]. Each definition that has a textbook number carries it. Each one is followed by a remark that says what it allows and forbids, an example, and a non-example. Most examples use the four-vertex graph H of Figure 1, which has the edges n_1n_2, n_1n_3, n_2n_3 and n_3n_4. Every classification of a sequence in these examples was checked by script [8].

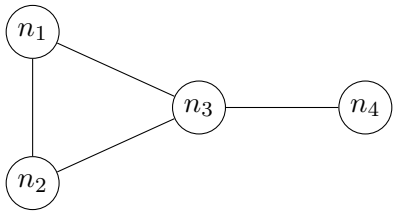


Figure 1: The example graph H: four towns, with roads n_1n_2, n_1n_3, n_2n_3 and n_3n_4. Its triangle n_1, n_2, n_3 gives it a loop trip, and n_4 hangs off n_3 by a single road.

3.1 Graph, adjacent vertices, adjacency matrix, degree

In plain words. A graph is a map of towns and roads. It is a list of the towns and a list of which pairs of towns have a road. A road joins two different towns. Between two towns there is at most one road.

Definition 3.1 (graph; textbook Definition 1.1.1)**.** A *graph* $G = (V, E)$ consists of a finite nonempty set V of *vertices* and a set E of *edges*, each edge being an unordered pair of different vertices; the edge with ends u and v is written uv [3, p. 9].

Remark 3.2. The textbook notes that a graph may be viewed as an antireflexive symmetric relation on V [3, p. 9]. Antireflexive means no edge joins a vertex to itself (no loops). Symmetric means uv and vu are the same edge. Because E is a set, two vertices have at most one edge between them. So the definition forbids loops, repeated edges and directions.

Example 3.3. H in Figure 1 has $V = \{n_1, n_2, n_3, n_4\}$ and four edges.

Non-example 3.4. A road map with a road from one town back to itself, or with two separate roads between the same two towns, is not a graph in this sense.

In plain words. Two towns are neighbours when a road joins them. That is all the next definition says.

Definition 3.5 (adjacent vertices; textbook Definition 1.1.4)**.** Vertices u and v are *adjacent* if $uv \in E$ [3, p. 11].

Remark 3.6. Adjacency is symmetric, and a vertex is never adjacent to itself, because there are no loops.

Example 3.7. In H the vertices n_3 and n_4 are adjacent.

Non-example 3.8. In H the vertices n_1 and n_4 are not adjacent: the edge n_1n_4 is not in E.

In plain words. The road table has one row and one column for each town. Put a one where the row town and the column town share a road. Put a zero everywhere else. The table tells you every road at a glance.

Definition 3.9 (adjacency matrix; textbook, after Definition 1.1.6)**.** Let $V = \{n_1, \ldots, n_N\}$. The *adjacency matrix* of G is the $N \times N$ matrix $A = (A_{ij})$ with $A_{ij} = 1$ if $e_{ij} \in E$ and $A_{ij} = 0$ otherwise [3, p. 11]. The board writes the same rule [1, 2]; see Convention 2.3 for its letters.

Remark 3.10. The only allowed entries are 0 and 1, so a repeated edge cannot be recorded. A 1 on the diagonal is forbidden because there are no loops. The textbook states the next two facts after the definition: $A_{ii} = 0$ for all i and $A_{ij} = A_{ji}$ for all i, j [3, p. 11]. The matrix depends on how the vertices are numbered: renumbering permutes the rows and the columns together.

Example 3.11. The graph H has the adjacency matrix

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The board's matrix $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ is the adjacency matrix of the graph on n_1, n_2, n_3 with the two edges n_1n_2 and n_2n_3 [1, 2]. We call that graph the *board graph*.

Non-example 3.12. The matrix $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ is not the adjacency matrix of a simple undirected graph: it is not symmetric.

In plain words. The degree of a town is the number of roads at that town. If you add up the roads at every town, you count each road twice, once from each end.

Definition 3.13 (degree; textbook Definition 1.2.1)**.** The *degree* $d(n_i)$ of a vertex n_i is the number of edges incident to n_i [3, p. 16].

Remark 3.14. The textbook introduces the degree after the adjacency matrix, so the row-sum property below is stated after this definition and not before it. The degree counts edges at the vertex, not vertices that can be reached from it.

Example 3.15. In H the degrees are $d(n_1) = 2$, $d(n_2) = 2$, $d(n_3) = 3$ and $d(n_4) = 1$ [8].

Non-example 3.16. In H the vertex n_4 can reach three other towns, but $d(n_4) = 1$, because only one road ends at n_4.

Theorem 3.17 (Euler; textbook Theorem 1.2.1)**.** *In every graph $G = (V, E)$ the degrees add up to twice the number of edges: $\sum_{v \in V} d(v) = 2|E|$ [3, p. 16].*

We use this theorem as the textbook states it, without proving it again.

Proposition 3.18 (properties of the adjacency matrix)**.** *Let A be the adjacency matrix of a simple undirected graph with N vertices.*

(1) A is symmetric: $A_{ij} = A_{ji}$.

(2) The diagonal is zero: $A_{ii} = 0$.

(3) The i-th row sum is the degree: $\sum_{j=1}^{N} A_{ij} = d(n_i)$. The column sums are the same, by symmetry.

(4) The sum of all entries of A is $2|E|$.

Proof. Properties (1) and (2) are the textbook's remark after the definition [3, p. 11]. For (1) the reason is that e_{ij} and e_{ji} name the same unordered pair. For (2) the reason is that an edge joins two different vertices.

(3) The row sum counts the indices j with $e_{ij} \in E$, that is, the neighbours of n_i. Every edge at n_i has the form e_{ij} with $j \neq i$, and two different edges at n_i have different other ends, because a simple graph has no repeated edge. So edges at n_i and neighbours of n_i match one to one, and the row sum equals the number of edges at n_i, which is $d(n_i)$.

(4) Add (3) over all i and use Theorem 3.17. □

For H the row sums of A_H are $2, 2, 3, 1$, which are the degrees, and the sum of all entries is $8 = 2 \cdot 4$ [8].

3.2 Walks, trails, paths, cycles, distance

In plain words. A trip follows roads from town to town. You may pass the same town twice. You may drive the same road twice. The number of drives is the size of the trip, and a road counts every time you drive it. Trips with fewer repeats get their own names.

Definition 3.19 (walk; textbook Definition 1.2.10)**.** A *walk of length k* from u_0 to u_k, also called a *(u_0, u_k)-walk*, is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges of G with $e_i = u_{i-1}u_i$ for $1 \leq i \leq k$ [3, p. 21].

Remark 3.20. In a walk, vertices and edges may repeat, and the length counts each edge as often as it occurs [3, p. 22]. The textbook abbreviates a walk by its vertex sequence $u_0, u_1, \ldots, u_k$. In a simple graph this loses nothing. The only possible edge e_i between u_{i-1} and u_i is the one edge $u_{i-1}u_i$. So a sequence of vertices is a walk exactly when every two consecutive vertices are adjacent, and the walk is determined by its vertex sequence. We write walks this way from now on. A walk of length 0 is a single vertex (Convention 2.6).

Example 3.21. In H the sequence $n_1, n_2, n_3, n_1, n_3, n_4$ is a walk of length 5: it drives n_1n_2, n_2n_3, n_3n_1, then n_1n_3 again, then n_3n_4. The sequence n_4, n_3, n_4 is a walk of length 2, and n_2 alone is a walk of length 0 [8]. The textbook gives its own example, a walk of length eight in its Fig. 1.2.5 that repeats vertices and edges [3, p. 22].

Non-example 3.22. In H the sequence n_1, n_4 is not a walk, because n_1 and n_4 are not adjacent. Neither is n_1, n_2, n_4.

In plain words. A round trip ends in the town where it started. It may repeat anything on the way.

Definition 3.23 (closed walk; textbook Definition 1.2.11)**.** A (u_0, u_k)-walk is *closed* if $u_0 = u_k$ [3, p. 22].

Remark 3.24. The definition forbids nothing else: a closed walk may repeat vertices and edges. A walk of length 0 is closed.

Example 3.25. In H the sequence n_1, n_2, n_1 is a closed walk of length 2.

Non-example 3.26. The walk n_1, n_2, n_3 is not closed, because it ends at n_3.

In plain words. Some trips never drive a road twice. They may still visit a town twice. The textbook calls such a trip a path. We avoid that word here, because it will mean something stricter. We say only that the trip never drives a road twice.

Definition 3.27 (trail; the textbook's "path", Definition 1.2.12)**.** A (u_0, u_k)-walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise different [3, p. 22]. The textbook calls it a (u_0, u_k)-path and measures its length by the number of its edges. By Convention 2.5 we call it a trail.

Remark 3.28. Vertices may repeat in a trail; edges may not. Every trail is a walk.

Example 3.29. In H the walk n_4, n_3, n_1, n_2, n_3 of length 4 is a trail. It visits n_3 twice, and it uses the four different edges n_4n_3, n_3n_1, n_1n_2 and n_2n_3 [8].

Non-example 3.30. The walk n_1, n_2, n_1 is not a trail, because it drives the road n_1n_2 twice. The length-5 walk of the example after Definition 3.19 is not a trail either: it uses n_1n_3 twice.

In plain words. The strictest trip never visits a town twice. In our course this is what the word path means. If you never visit a town twice, you also never drive a road twice.

Definition 3.31 (path; textbook Definition 1.2.13, "simple path")**.** A (u_0, u_k)-trail is a *path* (the textbook's *simple path*) if its vertices $u_0, \ldots, u_k$ are pairwise different [3, p. 22].

Remark 3.32. A path with k edges has $k+1$ different vertices. Every path is a trail, as Lemma 4.4 proves, so the definition only adds a restriction to Definition 3.27. This is the course meaning of path (Convention 2.5).

Example 3.33. In H the sequence n_1, n_3, n_4 is a path of length 2.

Non-example 3.34. In H the trail n_4, n_3, n_1, n_2, n_3 is not a path, because n_3 occurs twice.

In plain words. Often we care about two particular towns. We ask for a tidy route that starts in the first and ends in the second. This is the answer to question Q3, and Section 4 comes back to it.

Definition 3.35 (path between n_i and n_j; the course convention, Q3)**.** A *path between n_i and n_j* is a sequence $n_i = w_0, w_1, \ldots, w_k = n_j$ of pairwise different vertices with w_{t-1} adjacent to w_t for $1 \le t \le k$. It is a path in the sense of Definition 3.31, namely a simple (n_i, n_j)-path of the textbook [3, p. 22]. This reading is the course convention (Convention 2.5); it is not found on the board [1].

Remark 3.36. For $i \neq j$ the length k is at least 1. For $i = j$ the only path is the single vertex n_i, of length 0, because a sequence of length at least 1 from n_i to n_i repeats n_i. This is the reason the walk n_1, n_2, n_1 that the entry $(A^2)_{11}$ counts is not a path (Section 5.4).

Example 3.37. On the board graph, n_1, n_2, n_3 is a path between n_1 and n_3, of length 2.

Non-example 3.38. On the board graph, n_1, n_2, n_1, n_2, n_3 is a walk from n_1 to n_3 but not a path between them: it repeats n_1 and n_2 [8].

In plain words. A loop trip leaves a town and comes back to it. It never drives a road twice.

Definition 3.39 (cycle, simple cycle; textbook Definitions 1.2.14 and 1.2.15)**.** A (u_0, u_k)-trail is a *closed trail* or *cycle* if it is a closed walk, that is, $u_0 = u_k$ [3, p. 22]. A cycle is *simple* if only its first and last vertices repeat [3, p. 22].

Remark 3.40. The textbook's cycle is a closed trail, so it may pass through a vertex twice; the simple cycle is the stricter one. In a simple graph a closed trail of positive length has length at least 3. A closed trail of length 1 would be a loop, and one of length 2 would be u, w, u, which uses the edge uw twice. We prove nothing further about cycles. The board graph has no cycle of positive length: a trail in it has at most 2 edges, because it has only two edges, and a closed trail of positive length needs at least 3.

Example 3.41. In H the sequence n_1, n_2, n_3, n_1 is a simple cycle of length 3.

Non-example 3.42. The closed walk n_1, n_2, n_1 is not a cycle: it is not a trail. A cycle that is not simple needs two triangles that share a vertex, and H has only one triangle.

In plain words. The fewest-drives number between two towns is how far apart they are. If no trip joins them, they are infinitely far apart.

Definition 3.43 (distance; textbook Definition 1.2.16)**.** The *distance* $d_G(u, v)$ is the length of a shortest (u, v)-path of the textbook (a shortest trail in our words) if at least one exists, and $+\infty$ otherwise [3, pp. 22–23].

Remark 3.44. The distance is 0 exactly when $u = v$. Proposition 4.9 shows that it equals the least length of any walk, of any trail and of any path from u to v, so the choice of word does not change the number.

Example 3.45. In H the distance $d_H(n_4, n_1)$ is 2, along n_4, n_3, n_1 [8].

Non-example 3.46. In H the path n_4, n_3, n_2, n_1 joins n_4 to n_1 and has length 3, but 3 is not the distance, because a shorter path of length 2 exists.

4 Question Q3: paths between two vertices

Definition 3.35 answers Q3: a path between n_i and n_j is a sequence of pairwise different vertices from n_i to n_j in which consecutive vertices are adjacent. This section adds what the definition alone does not give. It compares the names that the course, the textbook and English texts use for the same objects. It then proves the four short results about walks, trails and paths on which Q1 and Q2 rest.

4.1 Three naming conventions side by side

In plain words. The word path is used for two different kinds of trip. One kind never drives a road twice. The other never visits a town twice. Books disagree on which kind gets the word. So a reader must always check which kind is meant.

Table 2 lists the objects of Definitions 3.19 to 3.39. The first column states the property, the second the name this note uses, the third the textbook's name with its number, and the fourth what English-language texts use where we checked.

The textbook's numbers come from [3, p. 22]. Two things limit the table. The word *trail* for Definition 1.2.12 is taken from a tertiary web page that we read only in part [9]; Grinberg defines walk and path but not trail [7]. And we did not check English usage of "closed trail" and "cycle", so those cells say so. A fourth reading exists: Bai's notes call a walk a path if it has no repeated vertex except possibly that the first and last vertices are equal [5, p. 1]. Under that reading a cycle counts as a path. We do not use it.

Table 2: The same objects under three conventions. The textbook's terms are given in English translation.

The sequence must satisfy	This note	Textbook (number)	English-language texts
consecutive vertices adjacent	walk	walk (1.2.10)	walk [7, Def. 2.9.1(a)]
a walk with $u_0 = u_k$	closed walk	closed walk (1.2.11)	not compared
no repeated edge	trail	"path" (1.2.12)	trail [9]
no repeated vertex	**path**	"simple path" (1.2.13)	path [7, Def. 2.9.1(c)]
a closed trail	cycle	"closed path" or "cycle" (1.2.14)	not compared
a cycle whose only repeated vertices are first and last	simple cycle	"simple cycle" (1.2.15)	not compared

Remark 4.1 (a danger for the reader). The textbook's definition of a connected graph (Definition 6.1 below) uses the textbook's "path", which is our trail. The course convention uses our path. Proposition 4.5 shows that the two readings describe the same connected graphs, so the choice does not change the answer to Q2. The choice would change other statements, for example a bound on the length.

4.2 Cutting loops out of a walk

In plain words. Suppose a trip visits some town twice. Between the two visits you drove around a loop. Cut the loop out. You still get from the start to the end, and the trip got shorter. Keep cutting until no town is visited twice. What is left is a tidy route.

Lemma 4.2 (every walk contains a path)**.** *Let u and v be vertices, equal or not. If there is a (u, v)-walk of length k, then there is a (u, v)-path of length at most k.*

The textbook proves this for $u \neq v$ by induction on the length of the walk and gives no bound on the length (Lemma 1.2.1, [3, p. 23]). Grinberg states the bound and covers $u = v$ [7, p. 39, Corollary 2.9.7]. We prove the strong form, so that no step depends on those sources.

Proof. We use strong induction on k.

Base case, $k = 0$. The walk is the single vertex u, and $u = v$. One vertex has pairwise different vertices, so it is a path of length $0 \leq 0$.

Induction hypothesis. Let $k \geq 1$ and suppose the statement holds for all walks of length less than k, between any two vertices.

Induction step. Let $w_0, \ldots, w_k$ be a (u, v)-walk. If its vertices are pairwise different, it is already a path of length k. Otherwise choose indices $i < j$ with $w_i = w_j$. Delete $w_{i+1}, \ldots, w_j$ to obtain $w_0, \ldots, w_i, w_{j+1}, \ldots, w_k$. Consecutive vertices are still adjacent: inside the first part and inside the last part nothing changed, and at the join $w_i = w_j$ is adjacent to w_{j+1}. If $j = k$ there is no join, and the sequence ends at $w_i = w_k = v$. The new sequence starts at u, ends at v, and has length $k - (j - i) < k$, because $j - i \geq 1$. By the induction hypothesis it contains a (u, v)-path of length at most $k - (j - i)$, hence at most k. □

Example 4.3. In H the walk $n_1, n_2, n_3, n_1, n_3, n_4$ has n_1 at positions 0 and 3. Deleting the vertices at positions 1 to 3 leaves n_1, n_3, n_4, a path of length 2. The walk $n_4, n_3, n_1, n_2, n_3, n_4$ has n_4 at both ends; cutting leaves the single vertex n_4, a path of length 0 [8].

Lemma 4.4 (a path is a trail)**.** *If the vertices of a walk are pairwise different, then its edges are pairwise different.*

In plain words. If you never visit a town twice, you cannot drive the same road twice.

Proof. Let $w_0, \dots, w_k$ have pairwise different vertices and suppose that two of its edges are equal: $\{w_{i-1}, w_i\} = \{w_{j-1}, w_j\}$ with $1 \le i < j \le k$. Then w_{i-1} equals w_{j-1} or w_j. The indices $j-1$ and j are both different from $i-1$, since $j-1 \ge i$. So two different positions hold the same vertex, which contradicts the assumption. □

In plain words. Three questions sound alike. Can you get from one town to another by any trip? By a trip that never drives a road twice? By a tidy route? The answers are always the same.

Proposition 4.5 (three readings of "joined")**.** *For vertices u and v of a simple graph the following are equivalent: (a) there is a (u,v)-walk; (b) there is a (u,v)-trail, that is, a (u,v)-path in the textbook's sense of Definition 1.2.12; (c) there is a (u,v)-path with pairwise different vertices.*

Proof. (c) implies (b) by Lemma 4.4. (b) implies (a), because a trail is a walk by Definition 3.27. (a) implies (c) by Lemma 4.2. □

Remark 4.6. By Proposition 4.5, the textbook's definition of a connected graph, which asks for a (u,v)-trail for every two vertices, holds exactly when every two vertices are joined by a path in the course sense. The board's definition read with Convention 2.5 and the textbook's definition therefore describe the same graphs. Grinberg proves only the equivalence of (a) and (c) [7, p. 42, Proposition 2.9.10]; the trail reading is proved here.

Example 4.7. On the board graph the walk n_1, n_2, n_1, n_2, n_3 shows that n_1 and n_3 are joined by a walk. Cutting the loop n_1, n_2, n_1 leaves the path n_1, n_2, n_3, which is also a trail [8].

In plain words. A tidy route visits each town at most once. So it cannot be longer than the number of towns minus one drives.

Proposition 4.8 (length bound)**.** *Let G have N vertices. (i) Every path in G has length at most $N-1$. (ii) If $u \neq v$ and some walk joins them, then a walk of length k with $1 \le k \le N-1$ joins them, and it can be chosen to be a path.*

The bound in (i) is also stated by Grinberg and by Gotti [7, p. 40], [6, p. 2].

Proof. (i) A path of length k has $k+1$ pairwise different vertices, all in V, so $k+1 \le N$. (ii) Apply Lemma 4.2 to the given walk. It yields a path of length at most that of the walk. The length is at least 1 because $u \neq v$, and at most $N-1$ by (i). A path is a walk. □

In plain words. How far apart are two towns? It is the fewest drives of any trip from one to the other. It does not matter whether we also forbid repeats, because forbidding repeats never lets a trip get shorter.

Proposition 4.9 (distance is the least walk length)**.** *If a (u,v)-walk exists, then $d_G(u,v)$ of Definition 3.43 equals the least length of a (u,v)-walk, the least length of a (u,v)-trail and the least length of a (u,v)-path, and it is at most $N-1$. If no walk exists, all three sets are empty and $d_G(u,v) = +\infty$.*

Proof. Let a, b and c be the least lengths of a walk, a trail and a path from u to v. Every path is a trail (Lemma 4.4) and every trail is a walk, so $a \le b \le c$. By Lemma 4.2 a shortest walk contains a path that is no longer, so $c \le a$. Hence $a = b = c$, and $d_G(u,v) = b$ by Definition 3.43. The bound $N-1$ comes from Proposition 4.8(i), applied to a shortest path. If a walk exists, a trail exists by Proposition 4.5, so the three sets are empty together. □

Remark 4.10 (the textbook's three properties of distance). The textbook lists three properties of the distance: it is ≥ 0 and equals 0 exactly when $u = v$; $d(u, v) = d(v, u)$; and $d(u, w) \leq d(u, v) + d(v, w)$ [3, pp. 22–23]. The reading of Proposition 4.9 gives each of them. A walk of length 0 exists from u only to u. Reversing a walk gives a walk of the same length in the other direction, because adjacency is symmetric. A shortest (u, v)-walk followed by a shortest (v, w)-walk is a (u, w)-walk of length $d(u, v) + d(v, w)$; if either term is $+\infty$ the inequality holds trivially.

5 Question Q1: what the entry $(A^k)_{ij}$ represents

Q1 asks what the entry in row i and column j of a power of A stands for. We read the mark on the board as an exponent k (Convention 2.2).

5.1 The theorem

In plain words. Start in one town and make a fixed number of drives. Each drive follows one road. You may turn round and go back along the road you just used. The table of roads, multiplied by itself that many times, tells you for every pair of towns how many different trips of that size lead from the first town to the second. It counts trips, not tidy routes. A trip may pass the same town twice.

Theorem 5.1 (walk counting)**.** *Let G be a finite simple undirected graph with vertices $n_1, \ldots, n_N$ and adjacency matrix A. For every integer $k \geq 1$ and all $i, j \in \{1, \ldots, N\}$, the entry $(A^k)_{ij}$ equals the number of walks of length k from n_i to n_j.*

The textbook has no such theorem (Section 1), so the proof below is our own. The statement is also found in the University of Galway notes (Theorem 1.1.3, [4, p. 4]), in Bai's notes (Theorem 1, [5, p. 1]) and in Gotti's notes (Proposition 2, [6, p. 1]). The method, induction on k with the walks split by their last-but-one vertex, is the one those notes use [4, 6].

Proof. We use induction on k. Let $W_k(i, j)$ denote the set of walks of length k from n_i to n_j, each written as a vertex sequence $(u_0, \ldots, u_k)$ with $u_0 = n_i$, $u_k = n_j$ and every two consecutive vertices adjacent (see the remark after Definition 3.19). The claim is $(A^k)_{ij} = |W_k(i, j)|$.

Base case, $k = 1$. A walk of length 1 from n_i to n_j is a sequence (n_i, n_j) with n_i and n_j adjacent. There is exactly one such walk if $e_{ij} \in E$ and none otherwise. So $|W_1(i, j)| = A_{ij} = (A^1)_{ij}$.

Induction hypothesis. Let $k \geq 1$ and suppose $(A^k)_{ir} = |W_k(i, r)|$ for all i and r.

Induction step. Take a walk $(u_0, \ldots, u_k, u_{k+1})$ in $W_{k+1}(i, j)$ and let $n_r = u_k$ be its last-but-one vertex; n_r is one of $n_1, \ldots, n_N$. Then $(u_0, \ldots, u_k)$ lies in $W_k(i, r)$, and n_r is adjacent to n_j. Conversely, any walk in $W_k(i, r)$ followed by n_j, where n_r and n_j are adjacent, is a walk in $W_{k+1}(i, j)$. So for each fixed r the walks in $W_{k+1}(i, j)$ with last-but-one vertex n_r match one to one the pairs consisting of a walk in $W_k(i, r)$ and the edge e_{rj}, where the edge exists exactly when $A_{rj} = 1$. Hence the number of such walks is $|W_k(i, r)|\, A_{rj}$, which is 0 when $A_{rj} = 0$. Every walk has exactly one last-but-one vertex, so the sets for different r are disjoint and together cover $W_{k+1}(i, j)$. Therefore

$$|W_{k+1}(i, j)| = \sum_{r=1}^{N} |W_k(i, r)|\, A_{rj} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj} = (A^{k+1})_{ij}.$$

The second equality is the induction hypothesis and the third is the definition of the matrix product. This completes the induction. □

Why the proof is valid. It uses three facts. A walk of length $k+1$ is a walk of length k plus one more edge. That edge exists exactly when the matrix entry is 1. The matrix product sums over the possible last-but-one vertex. Simplicity is used to write a walk as its vertex sequence. Nothing forbids a repeated vertex or a repeated edge, so the count is a count of walks. The product $(A^k)_{ir}A_{rj}$ multiplies a count by 0 or 1 and never checks which vertices the first part visited; this is why the proof cannot count paths.

Remark 5.2 (length zero). With $A^0 = I$, the entry $(A^0)_{ij}$ is 1 if $i = j$ and 0 otherwise. This matches the walks of length 0, which are the single vertices (Convention 2.6). The theorem is proved for $k \geq 1$, and the case $k = 0$ is that convention.

Proposition 5.3 (more properties of A^k)**.** *Let A be as in Theorem 5.1.*

(1) Every entry of A^k is a nonnegative integer.

(2) A^k is symmetric.

(3) $(A^2)_{ii} = d(n_i)$.

(4) The trace of A^2 (the sum of its diagonal entries) is $2|E|$.

Proof. (1) A has entries 0 and 1, and sums and products of nonnegative integers are nonnegative integers. (2) The transpose of a product reverses the order of the factors, so $(A^k)^T = (A^T)^k = A^k$. In walk terms, reversing a walk from n_i to n_j gives a walk of the same length from n_j to n_i, and this is a one-to-one correspondence. (3) $(A^2)_{ii} = \sum_r A_{ir}A_{ri} = \sum_r A_{ir}^2 = \sum_r A_{ir}$, because the entries are 0 or 1, and this equals $d(n_i)$ by Proposition 3.18(3). In walk terms, the walks of length 2 from n_i to itself are (n_i, n_r, n_i), one for each neighbour n_r. The Galway notes state the same fact [4, p. 4]. (4) The trace is the sum of the degrees, which is $2|E|$ by Theorem 3.17. □

5.2 A worked example: the board graph

In plain words. Take three towns in a row: the first has a road to the second, and the second has a road to the third. We write the road table and multiply it by itself. Then we list the trips behind a few of the numbers, to check that the table counts them.

The board graph has the edges n_1n_2 and n_2n_3 (Figure 2, left). Its powers, computed by script [8] and checked against an explicit enumeration of all walks for $k = 1, \ldots, 4$, are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

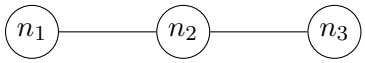
$$A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}, \quad A^4 = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 2 \end{pmatrix}.$$

The matrix A is symmetric with zero diagonal, and its row sums $1, 2, 1$ are the degrees; their sum is $4 = 2|E|$.

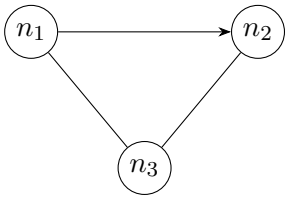
Table 3 lists the walks behind selected entries. Each count agrees with the matrix.

5.3 The board's drawing against the board's matrix

The board shows a drawing of three vertices beside its matrix [1, 2]. We read the drawing as a line from 1 to 2 with an arrowhead at 2, a line from 1 to 3, and a line between 3 and 2; this reading has medium confidence. The matrix encodes only the edges n_1n_2 and n_2n_3. Table 4 compares them pair by pair.



graph of the matrix



drawing as we read it

Figure 2: Left: the graph encoded by the board's matrix. Right: the board's drawing as we read it, with the arrowhead on the line from n_1 to n_2 and the extra line between n_1 and n_3. The reading of the drawing has medium confidence.

Table 3: Walks behind selected entries of the powers of the board matrix. All walks are written as vertex sequences. The last column lists those walks that are paths in the course sense.

Entry	Value	Walks counted	Paths among them
$(A^2)_{11}$	1	(n_1, n_2, n_1)	none
$(A^2)_{13}$	1	(n_1, n_2, n_3)	(n_1, n_2, n_3)
$(A^2)_{22}$	2	(n_2, n_1, n_2), (n_2, n_3, n_2)	none
$(A^2)_{12}$	0	no walk	none
$(A^3)_{12}$	2	(n_1, n_2, n_1, n_2), (n_1, n_2, n_3, n_2)	none
$(A^3)_{23}$	2	(n_2, n_1, n_2, n_3), (n_2, n_3, n_2, n_3)	none

Following Convention 2.4 the matrix is the example graph, and we do not hide the mismatch. For comparison, the graph of the drawing without the arrowhead is the triangle with edges n_1n_2, n_1n_3 and n_2n_3. Its powers [8] are

$$A_\triangle = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A_\triangle^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix},$$

$$A_\triangle^3 = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}, \quad A_\triangle^4 = \begin{pmatrix} 6 & 5 & 5 \\ 5 & 6 & 5 \\ 5 & 5 & 6 \end{pmatrix}.$$

Both graphs are connected, but the counts differ. For example $(A^2)_{13} = 1$ in both graphs, while $(A^2)_{11}$ is 1 for the board matrix and 2 for the triangle, which is the degree of n_1 in each. In the triangle $(A^3)_{11} = 2$ counts the closed walks (n_1, n_2, n_3, n_1) and (n_1, n_3, n_2, n_1), and $(A^3)_{12} = 3$ counts (n_1, n_2, n_1, n_2), (n_1, n_2, n_3, n_2) and (n_1, n_3, n_1, n_2), none of which is a path [8]. Which of the two graphs the lecturer meant cannot be settled from the photos.

5.4 Walks are not paths

In plain words. The table counts every trip, including a trip that goes to a neighbouring town and straight back. A tidy route visits each town at most once. So the table gives more than the number of tidy routes. It can also give a positive number when no tidy route exists.

Table 4: Drawing against matrix, pair by pair.

Pair	Drawing	Matrix	Agree
n_1n_2	line, arrowhead at n_2	$A_{12} = A_{21} = 1$	edge agrees; the arrowhead would make it directed, which a symmetric matrix does not
n_2n_3	line	$A_{23} = A_{32} = 1$	agree
n_1n_3	line	$A_{13} = A_{31} = 0$	disagree

Bai's notes say that the entry counts walks and not paths, because matrix multiplication accumulates all edge sequences, including those that revisit vertices [5, p. 1]. The board graph gives three concrete counterexamples to reading the entry as a number of paths; all three are in Table 3 and the powers above [8].

1. $(A^2)_{11} = 1 > 0$, but no path from n_1 to n_1 has length 2: a sequence (n_1, x, n_1) repeats n_1. The counted walk is (n_1, n_2, n_1). In general $(A^2)_{ii} = d(n_i) > 0$ whenever n_i has a neighbour (Proposition 5.3(3)).

2. $(A^3)_{12} = 2$, but exactly one path joins n_1 to n_2, namely the edge, of length 1, and no path has length 3. The two counted walks are (n_1, n_2, n_1, n_2) and (n_1, n_2, n_3, n_2).

3. A^4 is not the zero matrix although $N - 1 = 2$ and a path has at most $N - 1$ edges (Proposition 4.8). So the entries of high powers cannot be counts of paths.

Remark 5.4 (where the count equals a path count). For $k = 1$ every walk is a path. For $k = 2$ and $i \neq j$, the entry $(A^2)_{ij}$ is the number of paths of length 2 from n_i to n_j. The reason is that a walk (n_i, n_r, n_j) has distinct vertices exactly when $r \neq i$ and $r \neq j$, and this holds automatically, because n_r is adjacent to both and there are no loops. For $k \geq 3$ repeated vertices can occur even when $i \neq j$, as in the second counterexample.

Remark 5.5 (a misstatement in the literature). Theorem 1 of Paramadevan and Sotheeswaran says that the entry of A^l is the number of "different paths" of length l [10, p. 16]. As worded this is incorrect under any convention in which a path cannot revisit a vertex, by the first counterexample above. We cite that paper only for the statement of the connectivity test (Section 6), not for walk counting.

6 Question Q2: deciding connectivity from A

Q2 asks how to know from A whether the graph is connected. We first state what connected means in the textbook and show that it agrees with the board under the course convention. We then give a test that uses only A, prove it, and work two examples.

6.1 Connected graphs and components

In plain words. A group of towns is connected when you can get from any town to any other. If some town cannot be reached from another, the roads split the towns into separate groups. Call each such group an island. Towns on the same island can reach one another. Towns on different islands cannot.

Definition 6.1 (connected graph; textbook Definition 2.1.1)**.** A graph G is *connected* if for any two vertices u and v there is a (u, v)-path in G [3, p. 48]. The textbook's "path" here is Definition 1.2.12, a walk with different edges, which we call a trail (Convention 2.5). The board gives the same definition with the word path undefined [1].

Remark 6.2. The definition allows any route, even one that passes through a vertex twice, as long as no edge repeats. It says nothing about the length of the route, and it says nothing about A. By Proposition 4.5 it describes the same graphs as the reading with paths in the course sense, so the test below may use paths.

Example 6.3. The textbook's Fig. 2.1.1 shows connected graphs: a cycle, a 3×3 grid and a tree [3, p. 48]. The board graph is connected, and so is H.

Non-example 6.4. The textbook's Fig. 2.1.2 shows a graph that is not connected and has four connectivity components [3, p. 49]. A three-vertex graph whose only edge is n_1n_2 is not connected: no walk leads from n_3 to n_1.

In plain words. Being on the same island is a fair rule. Every town is on its own island. If one town can reach a second, the second can reach the first. If the first reaches the second and the second reaches a third, the first reaches the third. So the towns split into islands with no overlap.

Definition 6.5 (connectivity component; textbook, p. 49)**.** Write $u \sim v$ when a (u, v)-path exists in G. The textbook shows that this relation is reflexive, symmetric and transitive, hence an equivalence relation, so it splits V into classes $V_1, \ldots, V_p$. The subgraphs $G_1, \ldots, G_p$ induced by these classes are the *connectivity components* of G [3, p. 49].

Remark 6.6 (textbook Remark 2.1.1). A graph is connected if and only if it has exactly one component [3, p. 49]. The textbook's reflexivity ($v \sim v$) agrees with the convention that a single vertex is a walk of length 0 (Convention 2.6), but it does not by itself state that convention.

Example 6.7. The three-vertex graph with only the edge n_1n_2 has two components, with vertex sets $\{n_1, n_2\}$ and $\{n_3\}$. The board graph has one component.

Non-example 6.8. The sets $\{n_1\}$ and $\{n_3\}$ are not components of the board graph. The graph is connected, so its only component contains all three vertices.

6.2 The test

In plain words. The road table remembers how many trips of each size lead from one town to another. If that number is above zero for some size, a trip exists. A trip can always be trimmed to a tidy route by cutting out loops. A tidy route cannot use more drives than there are towns, less one. So we never need to look at trips bigger than that. Adding up the tables for all those sizes gives one table. A zero in it between two different towns means that no trip joins them.

For $N \geq 2$ define the matrices

$$B = A + A^2 + \cdots + A^{N-1}, \qquad R = (I + A)^{N-1}.$$

Here I is the identity matrix. B adds the walk counts of all lengths from 1 to $N - 1$. R is the power of "one plus the road table"; we show that its entries are positive combinations of the entries of B and of the identity.

Lemma 6.9 (pair criterion)**.** *Let $i \neq j$. There is a path from n_i to n_j if and only if $(A^k)_{ij} > 0$ for some k with $1 \leq k \leq N - 1$.*

Proof. ($\Rightarrow$) Let the path have k edges. Since $i \neq j$, $k \geq 1$. By Proposition 4.8(i), $k \leq N - 1$. The path is a walk of length k, so by Theorem 5.1 the entry $(A^k)_{ij}$, which counts such walks, is at least 1.

($\Leftarrow$) Suppose $(A^k)_{ij} > 0$. By Theorem 5.1 there is at least one walk of length k from n_i to n_j. By Lemma 4.2 it contains a path from n_i to n_j. □

Both directions need their own lemma. The direction from walk to path needs Lemma 4.2, because the entry counts walks and not paths (Section 5.4). The direction from path to walk needs only that a path is a walk, plus the length bound to limit k.

Theorem 6.10 (connectivity test)**.** *Let G be a simple undirected graph with $N \geq 2$ vertices and adjacency matrix A. The following are equivalent.*

(a) G is connected.

(b) For every $i \neq j$ there is $k \in \{1, \ldots, N-1\}$ with $(A^k)_{ij} > 0$.

(c) Every off-diagonal entry of $B = A + A^2 + \cdots + A^{N-1}$ is positive.

(d) Every entry of $R = (I + A)^{N-1}$ is positive.

The form (d) is stated by Gotti (Proposition 3, [6, p. 2]) and by Bai (Theorem 2, [5, p. 1]), and the form (c) by Paramadevan and Sotheeswaran (Theorem 2, [10, p. 17]). The textbook has neither (Section 1). Gotti's proof relies on the binomial expansion and the path bound. The proof in Bai's notes says the power "aggregates all paths", which is imprecise: the entries count weighted walks. Paramadevan and Sotheeswaran count "different paths" and pass from a nonzero entry to a path without the walk-to-path lemma. We therefore give a complete proof here; the equivalence of the forms (c) and (d) is our own.

Proof. (a) $\Longleftrightarrow$ (b). By Definition 6.1 and Proposition 4.5, (a) says that every pair of different vertices is joined by a path; pairs with $i = j$ are trivial. Lemma 6.9 turns each such pair statement into (b).

(b) $\Longleftrightarrow$ (c). By Proposition 5.3(1) every entry of every A^k is a nonnegative integer. A sum of nonnegative numbers is positive if and only if at least one term is positive. So $B_{ij} > 0$ if and only if $(A^k)_{ij} > 0$ for some k from 1 to $N-1$.

(c)$\Rightarrow$(d). The matrix I commutes with every matrix, so the binomial theorem gives

$$R = (I + A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \qquad A^0 = I.$$

Take a position (i, j) with $i \neq j$. The term $k = 0$ contributes $I_{ij} = 0$. Every other term is $\binom{N-1}{k}(A^k)_{ij}$ with $\binom{N-1}{k} \geq 1$ and $(A^k)_{ij} \geq 0$. By (c) and (b) some $k \in \{1, \ldots, N-1\}$ has $(A^k)_{ij} \geq 1$, so $R_{ij} \geq 1$. On the diagonal the term $k = 0$ contributes 1 and the other terms are at least 0, so $R_{ii} \geq 1$.

(d)$\Rightarrow$(c). Take $i \neq j$. Then $R_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k}(A^k)_{ij}$ is positive, and every term is nonnegative, so some term is positive. Hence some $(A^k)_{ij} > 0$, which is (b), and (b) is equivalent to (c) as shown. □

Remark 6.11 (one vertex). For $N = 1$ the graph is connected (the single vertex is joined to itself by the path of length 0, Convention 2.6), there are no off-diagonal entries, and $(I+A)^0 = I = (1)$ has no zero entry. So (a), (c) and (d) hold, and the equivalence is true for $N = 1$ as well.

Remark 6.12 (what the test allows and forbids). The test uses only A. It answers yes or no for the whole graph. It does not say which route to take. We note four limits.

(1) The single power A^{N-1} is not a valid test. In the board graph $(A^2)_{12} = 0$ although the graph is connected, because the only walks from n_1 to n_2 have odd length: every road goes between $\{n_1, n_3\}$ and $\{n_2\}$. The single power A misses the pair n_1, n_3 for the same reason [8]. The test must combine powers.

(2) The exponent $N-1$ cannot be lowered. For the path graph on N vertices (towns in a row) the least k with $(A^k)_{1N} > 0$ is $N-1$; the script gives $1, 2, 3, 4, 5, 6$ for $N = 2, \ldots, 7$ [8, 11].

(3) Adding the powers A^k with $k > N-1$ to B is allowed and unnecessary. All terms are nonnegative, so no positive entry can become zero.

(4) Only positivity matters. The entries of B and R grow quickly with N and are not themselves measures of connectedness.

6.3 Components and distance from the same matrices

In plain words. The same table also says which towns share an island. Two towns are on one island exactly when the table has a positive number for that pair. The first size at which the pair turns positive is the fewest drives between the two towns.

Corollary 6.13 (components from R)**.** *Let G have $N \geq 2$ vertices. Then n_i and n_j lie in the same component if and only if $R_{ij} > 0$. The component of n_i is the set of all n_j whose entry in row i of R is positive.*

Proof. For $i = j$ the vertices trivially lie in the same component, and $R_{ii} \geq 1$ as shown in (c)⇒(d) of Theorem 6.10. For $i \neq j$, the formula $R_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}$ shows that $R_{ij} > 0$ if and only if $(A^k)_{ij} > 0$ for some $k \in \{1, \ldots, N-1\}$, which by Lemma 6.9 holds if and only if a path joins n_i and n_j. This uses only the pair (i, j) and not the connectedness of the whole graph. Two vertices lie in the same class of the relation $\sim$ of Definition 6.5 exactly when a path joins them, by Proposition 4.5. □

Corollary 6.14 (distance from the powers)**.** *Let $i \neq j$ and suppose a path joins n_i to n_j. The least $k \geq 1$ with $(A^k)_{ij} > 0$ equals the distance $d_G(n_i, n_j)$.*

Proof. By Theorem 5.1, $(A^k)_{ij} > 0$ exactly when a walk of length k from n_i to n_j exists. So the least such k is the least length of a walk from n_i to n_j, and by Proposition 4.9 that is the distance. □

6.4 Two worked examples

In plain words. Take the three towns in a row. We add the road table to its own square. Every pair of different towns gets a positive number. So the towns are connected. Then we take three towns where only the first two have a road. The third town gets zeros, and the test says no.

Example 6.15 (connected: the board graph)**.** Let G be the board graph, with $N = 3$ and $N - 1 = 2$. Computed by script [8, 12],

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

$$B = A + A^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad R = (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

By Theorem 5.1 the entry $(A^2)_{13} = 1$ counts the one walk n_1, n_2, n_3; $(A^2)_{22} = 2$ counts n_2, n_1, n_2 and n_2, n_3, n_2; and $(A^2)_{12} = 0$, because a walk of length 2 from n_1 to n_2 would need a middle vertex adjacent to both, and none exists. Every off-diagonal entry of B is at least 1, and every entry of R is positive. By Theorem 6.10 the graph is connected, as the picture shows. As a check of the binomial formula, the entry $R_{13} = \binom{2}{1} A_{13} + \binom{2}{2} (A^2)_{13} = 2 \cdot 0 + 1 \cdot 1 = 1$ agrees with the printed matrix. The least k with $(A^k)_{ij} > 0$ is 1 for (n_1, n_2), 2 for (n_1, n_3) and 1 for (n_2, n_3), which are the distances (Corollary 6.14). For the triangle of Section 5.3 the same computation gives a matrix B with every entry 2 and a matrix R with every entry 3, so the test gives the same answer for both readings of the drawing [8].

Example 6.16 (not connected: one road only)**.** Let G' be the graph on n_1, n_2, n_3 with the single edge n_1n_2, so n_3 has no road. Computed by script [8, 12],

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The off-diagonal entries $B_{13}, B_{23}, B_{31}, B_{32}$ are 0, and so are the same positions of R. By Theorem 6.10 the graph is not connected. By Corollary 6.13 the components are $\{n_1, n_2\}$, because row 1 of R is positive at positions 1 and 2, and $\{n_3\}$, because row 3 is positive only at position 3. That is two components, as the picture shows. No k in $\{1, 2\}$ makes $(A^k)_{13}$ or $(A^k)_{23}$ positive. The diagonal of A^2 is $(1, 1, 0)$, the degrees; a positive diagonal entry does not mean the graph is connected.

6.5 Breadth-first search as an alternative

In plain words. There is another way to find out. Stand in one town. Visit its neighbours. Then visit their neighbours, and keep going, never visiting a town twice. When you run out of new towns, count the towns you saw. If you saw all of them, the towns are connected. If some town never turned up, they are not.

Breadth-first search from a start vertex s keeps a queue and a list of discovered vertices. It takes a vertex v from the queue, scans the neighbours of v (on the adjacency matrix: the positions with a 1 in row v of A) and puts each undiscovered neighbour into the queue. A lecture at the University of Waterloo proves that every discovered vertex has a walk from s, and that every vertex with a walk from s is discovered; together, there is a walk from s to v if and only if v is discovered [13, slides 8 to 10]. The same lecture lists testing connectivity as an application, with the condition that all vertices are connected to s, but it gives no proof of that step [13, slide 10]. We supply it.

Proposition 6.17 (connectivity by search)**.** *A simple undirected graph G is connected if and only if breadth-first search from any one vertex s discovers every vertex.*

Proof. Suppose G is connected. Each vertex v has a path, hence a walk, from s, so v is discovered. Conversely suppose all vertices are discovered, and take any u and v. There are walks from s to u and from s to v. Reading the first backwards gives a walk from u to s, because adjacency is symmetric in an undirected graph. Attaching the second gives a (u, v)-walk, which contains a path by Lemma 4.2. So G is connected. □

Remark 6.18 (cost; an estimate)*.* Counting the work from the algorithm as described, each vertex is dequeued at most once and each dequeue scans one row of N entries, so the search reads at most about N^2 matrix entries. Computing the $N - 2$ products of the sum form by the definition of matrix multiplication costs about N^3 each, so about N^4 in total. This is our own estimate and not a result of a source; the lecture's bound of order $n + m$ is for adjacency lists, not for a matrix [13, slide 7]. Complexity is outside the scope of this note beyond this remark. The search also gives components and distances, which the matrix test gives only through R (Corollaries 6.13 and 6.14).

6.6 A finite check of the test

The test and its proof were also checked by computer on every labelled simple graph with $N = 2, \ldots, 6$ vertices (2, 8, 64, 1024 and 32768 graphs, of which 1, 4, 38, 728 and 26704 are connected). For each graph the sum test and the $(I + A)^{N-1}$ test were compared with a union-find connectivity check from the edge list that uses no matrices. For $N \leq 5$ the walk counts were also compared with an explicit enumeration, and the least k with a positive entry was compared with the breadth-first distance. All of them agreed, with no disagreement found [11]. This is an experimental check on small cases and not a proof; the proof is Theorem 6.10.

7 Discussion: limits and competing readings

- **Scope.** Every result is for finite simple undirected graphs (Convention 2.1). With loops the diagonal entries of A would need a convention and the row-sum property would change. With repeated edges the entries would be edge multiplicities, a vertex sequence would no longer determine a walk, and the counting theorem would count walks with a choice of edge. We did not work these cases out.

- **What the board leaves open.** The mark above A, the arrowhead and the extra line in the drawing are readings, not facts (Convention 2.2, Section 5.3). The photos cannot say what the lecturer meant by them. If the mark were a prime or quotation marks, Q1 would ask about another matrix; the assignment text rules that out, but the board alone does not.

- **The meaning of "path".** The board does not define it. The result of Q2 does not depend on the choice between the textbook's trail reading and the course's no-repeated-vertex reading (Proposition 4.5), but other statements do, for example the bound $N - 1$ on the length (Proposition 4.8), which holds for paths and not for trails.

- **Length zero.** Whether the textbook admits walks of length 0 is not settled by the pages we viewed. We adopted it as a convention (Convention 2.6). The test itself uses only pairs of different vertices.

- **Sources.** The three sources that state the counting theorem and the three that state the test are lecture notes or a small journal article; the Galway notes carry no author and no date, and we did not read Bai's notes in full. None is a research source. The theorem and the test are standard, and the proofs here do not depend on the proofs in those sources. The textbook was read in part, so the statement that it lacks the theorem and the test is the result of a search and of the pages viewed (Section 1), not of a full read. The word *trail* rests on a tertiary web page (Table 2).

- **Competing explanation of the sum.** Two sources explain the sum test as "the sum counts all paths of length at most $N - 1$" [5, 10]. As a statement about counts this is wrong: the entries count walks, weighted by binomial coefficients in R. As a statement about positivity it is right, and only through the walk-to-path lemma.

- **Size of the numbers.** Entries of A^k and R grow quickly: A^4 of the board graph already has an entry 4. Only positivity matters for the test. We make no claim about growth rates, which would need eigenvalue methods and lie outside the scope.

- **Checks.** The computations in this note are checks of examples and of small cases; the proofs carry the results.

8 Conclusion

The three questions have direct answers.

- **Q1.** For a simple undirected graph with adjacency matrix A and $k \geq 1$, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j (Theorem 5.1). It is not the number of paths. In the board graph $(A^2)_{11} = 1$, although no path leads from n_1 to itself with two edges, and A^4 is not zero although a path has at most 2 edges (Section 5.4).
- **Q2.** A graph with $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 6.10). The proof uses the walk-counting theorem, the fact that a walk contains a path (Lemma 4.2) and the bound $N - 1$ on the length of a path (Proposition 4.8). The board matrix passes the test and is connected; the graph with the single edge n_1n_2 fails it and has components $\{n_1, n_2\}$ and $\{n_3\}$ (Section 6.4).
- **Q3.** A path between n_i and n_j is a sequence of pairwise different vertices from n_i to n_j in which consecutive vertices are adjacent (Definition 3.35). It is the textbook's simple path, Definition 1.2.13 [3, p. 22]. The textbook's own word "path" means a trail, and Proposition 4.5 shows that this difference does not change which graphs are connected.

The board's drawing and matrix disagree on one edge, and we followed the matrix, as the problem statement on the board says the graph is given through its adjacency matrix (Convention 2.4).

A Sources, pages checked and how to rerun the computations

Sources and how much of each was read

- The course textbook [3]: read in part. We viewed printed pages 7, 11 and 16, pages 21 to 23, and pages 48 and 49 as images, and page 9 through its text layer, and we searched the text for the adjacency matrix. Printed page numbers are the PDF page minus one. The text layer drops most mathematical symbols, so formulas were taken from the page images.
- The two whiteboard photos [1, 2]: read together as one board.
- The Galway notes [4], Gotti's notes [6] and the journal article of Paramadevan and Sotheeswaran [10]: read in full. Bai's notes [5], Grinberg's book [7] and the Waterloo slides [13]: read in part, at the places cited. The Wikipedia page [9]: read in part, through a summarising tool, so its wording is as returned and not checked against the raw page.

All web sources were accessed on 2 November 2026 (see the bibliography).

The computations

Every matrix in the text, every walk count and every classification of an example sequence comes from one script, `appendices/scripts/paper_computations.py` [8], run from the paper folder with

```
python3 -I appendices/scripts/paper_computations.py \
    > appendices/scripts/paper_computations_output.txt
```

It writes `appendices/generated-matrices.tex`, which this document inputs, so no matrix is typed by hand. It computes powers by the definition of the matrix product and checks every entry of A^k for $k = 1, \ldots, 4$ against an explicit enumeration of walks. The script

appendices/scripts/m3_examples.py [12] computes the two three-vertex examples independently, and appendices/scripts/m3_exhaustive_check.py [11] performs the finite check of the test on all graphs with up to six vertices. The script appendices/scripts/m2_paths.py cuts every walk of length at most 6 in all 1099 labelled simple graphs on at most five vertices and found no violation of Lemma 4.2; its output is saved beside it. Run each with python3 -I.

References

[1] *Lecture whiteboard photo 1.* Local file sources/whiteboard-1.jpg. Photo of the lecture board, supplied with the assignment; author and date not stated. n.d.

[2] *Lecture whiteboard photo 2.* Local file sources/whiteboard-2.jpg. Second photo of the same board, supplied with the assignment; author and date not stated. n.d.

[3] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory (Grafneri tesutyun), study guide.* Yerevan State University Press, Yerevan, in Armenian; local file Grafneri_tesutyun.pdf. Course textbook, supplied with the assignment. Page numbers in this paper are the printed page numbers. 2015.

[4] *Matrices and Graphs, 1.1 The Adjacency Matrix.* University of Galway linear algebra notes, maths.nuigalway.ie, file section1-1.pdf. n.d. URL: https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf (visited on Nov. 2, 2026).

[5] S. Bai. *Important Properties and Applications of the Adjacency Matrix (20250310).* BIMSA lecture notes. 2025. URL: https://bimsa.net/doc/notes/48535.pdf (visited on Nov. 2, 2026).

[6] F. Gotti. *MIT 18.211: Combinatorial Analysis, Lecture 27: Adjacency Matrices and The Matrix-Tree Theorem.* MIT course notes, math.mit.edu/~fgotti. n.d. URL: https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/27.%20The%20Matrix-Tree%20Theorem/The%20Matrix-Tree%20Theorem.pdf (visited on Nov. 2, 2026).

[7] D. Grinberg. *An introduction to graph theory.* arXiv:2308.04512, version of June 7, 2025. 2025. URL: https://arxiv.org/abs/2308.04512 (visited on Nov. 2, 2026).

[8] *paper_computations.py: every matrix, count and classification printed in the paper.* Local file appendices/scripts/paper_computations.py. Pure Python script written for this paper. n.d.

[9] *Path (graph theory).* Wikipedia, English edition. n.d. URL: https://en.wikipedia.org/wiki/Path_(graph_theory) (visited on Nov. 2, 2026).

[10] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *Journal of Science of the Eastern University, Sri Lanka (JSc EUSL)* 12.1 (2021), pp. 13–21. URL: https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf (visited on Nov. 2, 2026).

[11] *m3_exhaustive_check.py: connectivity test checked on all labelled graphs with 2 to 6 vertices.* Local file appendices/scripts/m3_exhaustive_check.py. Pure Python script written for this paper. n.d.

[12] *m3_examples.py: matrices of the connected and disconnected three-vertex examples.* Local file appendices/scripts/m3_examples.py. Pure Python script written for this paper. n.d.

[13] E. Schost. *CS 341: Algorithms, Lecture 10: Graphs, breadth first search.* University of Waterloo, Cheriton School of Computer Science, lecture slides. n.d. URL: https://www.student.cs.uwaterloo.ca/~cs341/lectures/lecture10.pdf (visited on Nov. 2, 2026).